

## 2b. Content of Computer systems (J277/01)

| 1.1 – Systems architecture                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                          |
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| Sub topic                                                                                                                                                                                                               | Guidance                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>1.1.1 Architecture of the CPU</b>                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                          |
| <input type="checkbox"/> The purpose of the CPU: <ul style="list-style-type: none"><li>○ The fetch-execute cycle</li></ul>                                                                                              | <b>Required</b> <ul style="list-style-type: none"><li>✓ What actions occur at each stage of the fetch-execute cycle</li><li>✓ The role/purpose of each component and what it manages, stores, or controls during the fetch-execute cycle</li><li>✓ The purpose of each register, what it stores (data or address)</li><li>✓ The difference between storing data and an address</li></ul> |
| <input type="checkbox"/> Common CPU components and their function: <ul style="list-style-type: none"><li>○ ALU (Arithmetic Logic Unit)</li><li>○ CU (Control Unit)</li><li>○ Cache</li><li>○ Registers</li></ul>        |                                                                                                                                                                                                                                                                                                                                                                                          |
| <input type="checkbox"/> Von Neumann architecture: <ul style="list-style-type: none"><li>○ MAR (Memory Address Register)</li><li>○ MDR (Memory Data Register)</li><li>○ Program Counter</li><li>○ Accumulator</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                                                                                                                                                                                         | <b>Not required</b> <ul style="list-style-type: none"><li>✗ Knowledge of passing of data between registers in each stage</li></ul>                                                                                                                                                                                                                                                       |